Shipping-document extraction: zero-shot VLM + second-reader review flag

Repo: (repo URL pending) · test predictions: `test_predictions.json` · notes behind every decision: `analysis-notes/01–12` · No HF model / W&B run: nothing is trained.

1. Approach and why

```
pages → classify p1 → Qwen3-VL-8B (4-bit, MLX), per-type prompt, all pages in one call → column roles + 2 repairs → normalise
                                          PaddleOCR-VL reads each page → "value not on the page" → review flags
```

- **Zero-shot VLM, not OCR + rules.** In a 40-doc bake-off covering all 18 suppliers (note 09), Qwen3-VL-8B read supplier part numbers exactly 98.9% (digital) / 92.3% (scanned) of the time, above the *ceiling* of PP-OCRv5 (95.1 / 86.0: a value OCR never emits can't be extracted). It got 100% of header fields and 0 of 15 false fills on illegible fields with no rules at all.
- **No fine-tuning.** Train has 18 suppliers whose ~40-part catalogues are reused in 99% of dev rows (note 02), so a tuned model would learn to recall catalogues, which is exactly what fails on unseen suppliers. It would also cost a GPU we chose not to rent.
- **Field meaning from the model, formats from code.** The invoice prompt asks the model to name each column's *role* from the seller's point of view ("Your/Customer" = buyer) before it fills rows. This targets the costliest error (one swapped column zeroes every row). Code then does dates (ISO, ambiguous day/month flagged), numbers (incl. European `1.234,50`), currency and IATA codes. No rule mentions a supplier, prefix or layout position.
- **Sized for 24 GB.** Size sweep (note 11): 2B 55.5 OVERALL with false fills, 4B 84.0, 8B 4-bit 90.5, 8B 8-bit 89.4 (40-doc bake-off). With the final prompt, **8B 4-bit** scored 91.0 vs 90.3 for 8-bit, with 0 vs 1 false fill, at 19 vs 31 s/doc and a 7.5 vs 11.9 GB peak. We kept 4-bit, which leaves about two-thirds of the 24 GB free. The quantisation risk is real but not decisive: both widths read `PO` as `P0` on one invoice.
- **Null only when obscured.** Null on a readable field scores like a wrong value (note 07), so readable-but-uncertain values are kept and *flagged*; only absent, blacked-out or scribbled values are null.

2. Dev results (100 docs; official scorer, identical numbers from `eval.evaluate`)

slice	docs	OVERALL	header %	row F1	docs exact %	false fills
all	40	**91.0**	100.0	92.5	70.0	0/15
scanned	22	**89.6**	100.0	92.1	63.6	0/9
digital	18	**92.7**	100.0	92.9	77.8	0/6
multi-page	13	**82.9**	100.0	91.9	30.8	0/5
repeated parts	10	**86.9**	100.0	87.2	60.0	0/1
illegible field	6	**88.4**	100.0	87.8	66.7	0/8
waybills	8	**100.0**	100.0	–	100.0	0/1
invoices	32	**89.5**	100.0	92.5	62.5	0/14
unseen: fold 4, clean (no prompt change from these)	8	**84.8**	100.0	87.1	50.0	0/5
seen: other folds (prompt iterated on errors here)	32	**92.9**	100.0	94.8	75.0	0/10

3. Failure modes

(pending)

4. A new supplier layout

- **Nothing to retrain.** The prompt describes field *meanings* (seller vs buyer part number, PO even when it looks like a part number, totals only from the labelled total), not layouts; header wordings appear only as examples. A new layout goes through as-is, and fold 4 measures exactly this.
- **Monitor, then fix generically.** Review-flag rates per supplier show a new layout that reads badly. A fix is a meaning rule or a repair that holds for any layout (as the two existing column repairs do), checked on a held-out fold before it ships, never a supplier-specific rule.
- **Volume.** At ~10k docs/day this design moves to a 24 GB GPU with vLLM batching and JSON-schema-constrained decoding (note 10); that path is a recommendation, untested here.

5. Needs-review flag (separate file; never changes a value)

A value is flagged when the independent reader (PaddleOCR-VL) does not find it on the page, or when a slash date could be day- or month-first; a whole document is flagged if the output was truncated. On dev: **document level** precision 68.8%, recall 91.7% (a flagged doc is not fully correct); **value level** precision 39.4%, recall 90.3% (header fields and predicted rows). Column-mapping errors are invisible to a text check by design, so they are only caught at document level when another value is also flagged.

```
doc    precision  68.8%  recall  91.7%  (flagged 16, errors 12, caught 11)
value  precision  39.4%  recall  90.3%  (flagged 71, errors 31, caught 28)
  not_in_ocr                              flags   71  precision  39.4%  errors caught  28 (only by this signal 28)
```

6. Limits and honesty

- I looked at all 18 train suppliers while analysing the data, and prompt v2 was written after reading dev errors (fold 4 excluded). Dev numbers are therefore optimistic. The fold-4 row is the cleanest unseen estimate, but it has only 8 docs.
- Throughput on the Mac: over 51 docs, 22 s/doc median, 93 s p95, 94 docs/hour on one Mac, 8.1 GB peak for the extractor (+ PaddleOCR-VL ~3 s/page, 2.3 GB, run separately). The transformers backend (`--backend hf`) for graders without a Mac is implemented but was not run.
- Cut for time: per-fold re-runs, scan upscaling, a second OCR tier for high-precision flags.